

IS-02: A counterexample to the spectral-uniqueness locus condition

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Status of this manuscript: Independently checked resolution (Codex-agent review).

Outcome: Negative resolution.

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The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently checked the complete proof against the exact repository target on 11 September 2026; its [detailed review](#) records **PASS**. The theorem and proof text below are unchanged from the reviewed submission. This verification is an independent agent review, not external human peer review or a formal proof certificate.

Theorem IS-02: proposed counterexample

For real symmetric entrywise nonnegative stochastic matrices, define spectral uniqueness to mean uniqueness up to simultaneous permutation of rows and columns among matrices with the same spectrum, including multiplicity. Put

$$C_n = \frac{\mathbf{1}\mathbf{1}^T - I_n}{n-1}.$$

The proposed necessary condition places every spectrally unique matrix with positive trace, for $n \geq 4$, in the union of $[I_n, C_n]$ and the segments $[I_n, V]$ and $[C_n, V]$ over vertices V of the symmetric stochastic polytope.

Claim. The matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}$$

is a counterexample.

1. Spectral uniqueness

The matrix is symmetric, nonnegative and stochastic. Its spectrum is $\{1, 1, 0, -1\}$ and its trace is one. Let B be another real symmetric nonnegative stochastic matrix with this spectrum. For every real x ,

$$x^T(I - B)x = \frac{1}{2} \sum_{i,j} b_{ij}(x_i - x_j)^2.$$

The kernel consists of vectors constant on each connected component of the graph with an edge ij when $i \neq j$ and $b_{ij} > 0$. Thus the multiplicity of the eigenvalue one is the number of components, and B has exactly two components.

Any connected component carrying the eigenvalue -1 has even order. To see this, restrict a nonzero real eigenvector z with eigenvalue -1 to such a component. Then

$$0 = z^T(I + B)z = \frac{1}{2} \sum_{i,j} b_{ij}(z_i + z_j)^2.$$

Along any edge, $z_i = -z_j$. Connectivity implies that all coordinates have the same nonzero magnitude. The component is therefore bipartite, has no positive diagonal entries, and has no edges within either sign class. The total weight crossing from a sign class to the other is

its number of vertices, by stochasticity. Symmetry makes the two crossing totals equal, so the two classes have equal cardinality.

There are four vertices and two nonempty components. The component carrying -1 must consequently have order two, and the other component also has order two. Every symmetric stochastic matrix of order two is

$$B(t) = \begin{pmatrix} t & 1-t \\ 1-t & t \end{pmatrix}, \quad \sigma(B(t)) = \{1, 2t-1\}.$$

The two blocks of B must be $B(0)$ and $B(1/2)$. Therefore B is permutation-similar to A , proving spectral uniqueness.

2. Exclusion from the asserted locus

Let $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. The identity

$$A = \frac{1}{2} \text{diag}(S, I_2) + \frac{1}{2} \text{diag}(S, S)$$

is a nontrivial convex combination of two distinct symmetric stochastic matrices. Hence A is not a vertex.

If $A = (1-t)I_4 + tV$ with $0 \leq t \leq 1$ and V a vertex, the zero entry A_{11} and nonnegativity force $t = 1$. This would give $A = V$, which is impossible.

If $A = (1-t)C_4 + tV$, the zero entry A_{13} , together with $(C_4)_{13} = 1/3$ and $V_{13} \geq 0$, again forces $t = 1$ and the same contradiction.

Finally, writing $A = (1-t)I_4 + tC_4$, its zero $(1,3)$ entry forces $t = 0$, whereas $A \neq I_4$. This excludes the remaining segment and completes the proposed negative resolution. $\square$

Scope and review notes

The counterexample has $n = 4$, which is an allowed dimension. It uses entrywise nonnegativity, symmetry, row sums one, positive trace, and spectral uniqueness with multiplicities, exactly as in the entry. A counterexample in one allowed dimension refutes the universal necessary condition; a classification in all dimensions is not claimed.

The independent review checked the component argument for spectral uniqueness and the distinction between vertices of the symmetric stochastic polytope and permutation matrices. The proof does not assume those two vertex notions coincide.

The packaging pass checked the repository statement and contribution rules on 11 September 2026. It did not conduct a new exhaustive literature search or establish novelty.

Source references

Alex Townsend, *Open Problems in Numerical Linear Algebra* (2026), [entry IS-02](#), repository snapshot b412319, accessed 11 September 2026.

Mourad and Abbas, Conjecture 5.1, p. 10 of the [2013 manuscript](#); [published article](#), *Linear and Multilinear Algebra* 63 (2015), 869–881. These are the original-source locators given by the catalog.